



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

Semester: SUMMER 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 01

Experiment Name: Verification of KVL and KCL

Group No: 01

Group members:

<i>Sl.</i>	<i>Name</i>	<i>ID</i>
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5.		
6. (if applicable)		

Date of Submission: 22/06/2026

Prepared by: Ruhinul Haque Bhuiyan

3. Equipment :

- One multimeter
- Three Resistors
- One DC power Supply

4. Circuit Diagram :

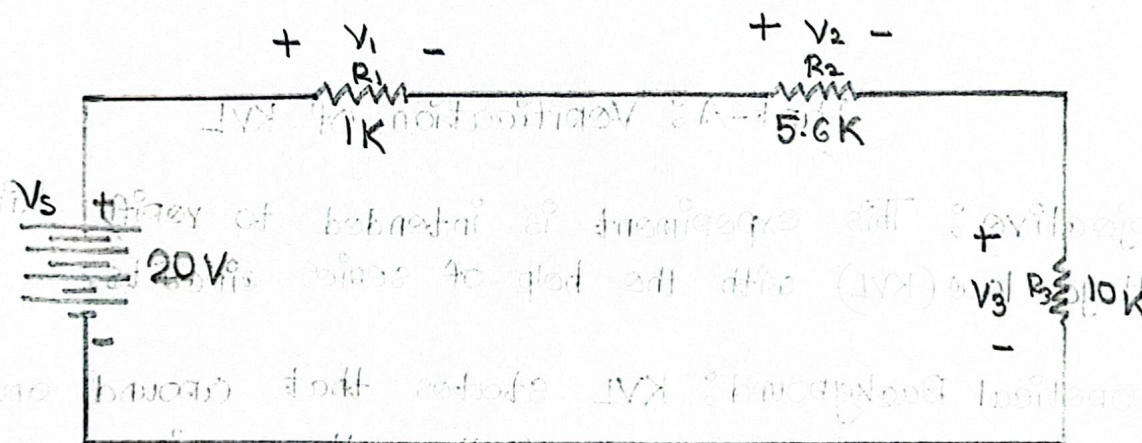


Fig 1 : Series circuit to demonstrate KVL

5. Data Table :

Verification of KVL :

V_s (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1 + V_2 + V_3$ (V)
19.97	1.20	6.68	12.08	19.96

6. Lab Report Questions :

a) Ans :

Voltmeter Connection Rule $\Rightarrow$ A voltmeter must always be connected in parallel across the component whose voltage drop is being measured.

Ammeter Connection Rule $\Rightarrow$ An ammeter must always be connected in series within the specific branch where current needs to be measured.

b) Ans :

The electrical power dissipation (P) of a resistor can be calculated using Ohm's Law and Joule's Law through any of the following formulas :

$$P = V \cdot I$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

Where,

V = Voltage drop across resistor

I = Current through resistor

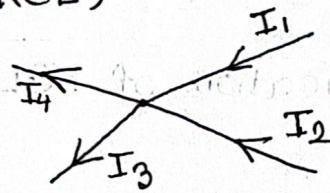
R = Ohmic Value of resistor

Part-B: Verification of KCL

1. Objective: This experiment is intended to verify Kirchhoff's current law (KCL) with the help of a series-parallel circuit.

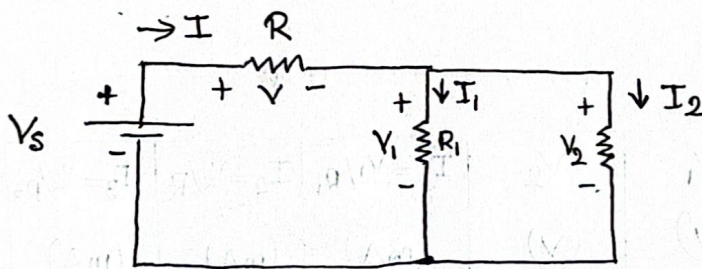
2. Theoretical Background: KCL states that the algebraic sum of the currents entering any node equals the sum of the currents leaving the node.

According to KCL,



$$\Rightarrow I_1 + I_2 = I_3 + I_4$$

$$\sum I_{\text{ent}} = \sum I_{\text{ext}}$$



$$I = I_1 + I_2$$

$$\Rightarrow I = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$\Rightarrow \frac{V}{R} = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

3. Equipment:

- Three resistors
- One multimeter
- One DC supply

4. Circuit Diagram:

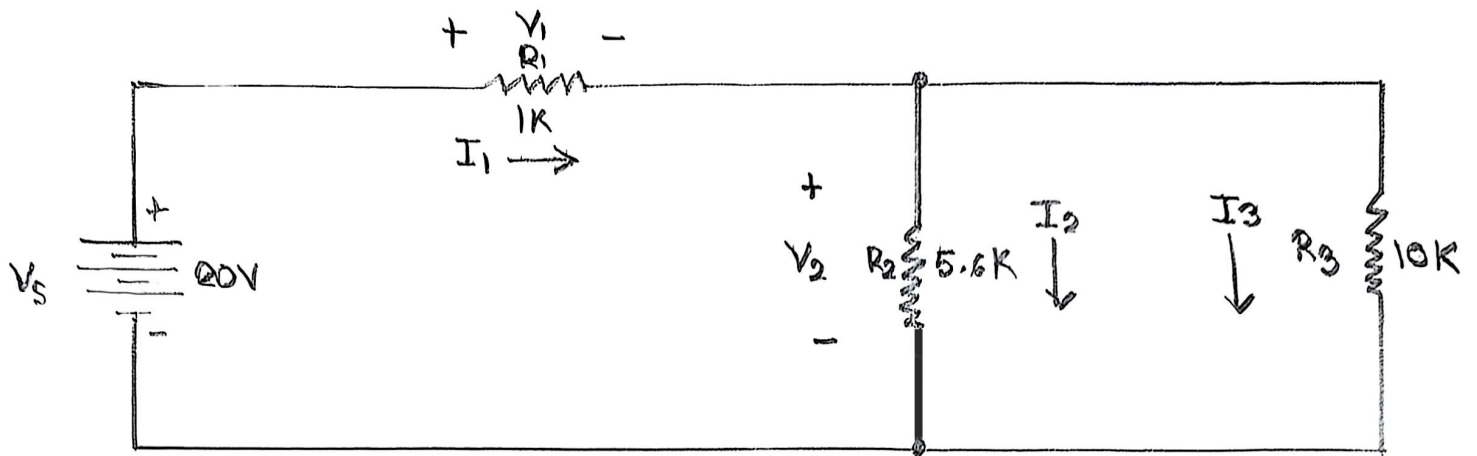


Fig. 2: Verification of KCL

5. Data Table:

Verification of KCL:

R_1 (K Ω)	R_2 (K Ω)	R_3 (K Ω)	V_1 (V)	V_2 (V)	$I_1 = V_1/R_1$ (mA)	$I_2 = V_2/R_2$ (mA)	$I_3 = V_2/R_3$ (mA)	$I_2 + I_3$ (mA)
1	5.5	9.96	4.43	15.68	4.43	2.85	1.57	4.42

6. Lab Report Questions :

a) Ans :

We know the formula : $P = V \cdot I$

Now,

Total Power Supplied by the DC Source (P_s) :

$$\begin{aligned} P_s &= V_s \cdot I_1 \\ &= 20.1 \text{ V} \times 4.43 \text{ mA} \\ &= 89.04 \text{ mW} \end{aligned}$$

Power Absorbed by R_1 :

$$\begin{aligned} P_{R_1} &= V_1 \cdot I_1 \\ &= 4.43 \text{ V} \times 4.43 \text{ mA} \\ &= 19.62 \text{ mW} \end{aligned}$$

Power Absorbed by R_2 :

$$\begin{aligned} P_{R_2} &= V_2 \cdot I_2 \\ &= 15.68 \text{ V} \times 2.85 \text{ mA} \\ &= 44.69 \text{ mW} \end{aligned}$$

Power Absorbed by R_3 :

$$\begin{aligned} P_{R_3} &= V_2 \cdot I_3 \\ &= 15.68 \text{ V} \times 1.57 \text{ mA} \\ &= 24.62 \text{ mW} \end{aligned}$$

b) Ans :

Yes.

Supplied Power : 89.04 mW

Absorbed Power : 88.93 mW $\approx$ 89.04 mW

EEE101L

Summer-2026

Group: 01

Name: i) Ruhinul Haque Bhuiyan

ii) Manatin Tasbik Munshed

iii) Amit Paul

iv) Dibyendu Roy Dibya

Data Table

Verification of KVL

V_s	V_1	V_2	V_3	$V_1 + V_2 + V_3$
19.97V 19.97V 20V	1.20V	6.68V	12.08	19.96V

Verification of KQI

R_1 (K Ω)	R_2 (K Ω)	R_3 (K Ω)	V_1 (V)	V_2 (V)	$I_1 = V_1/R_1$ (mA)	$I_2 = V_2/R_2$ (mA)	$I_3 = V_3/R_3$ (mA)	$I_2 + I_3$ (mA)
1	5.5	9.96	1.20 4.43	18.90 15.68	1.20 4.43	3.11 2.85	1.90 1.57	5.84 4.42